Jashinder **Kumar** | Electrical & Embedded Systems Engineering Student

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Summary

Dedicated Electrical Engineering student specializing in embedded system hardware, power electronics conversion, and microcontrollers. Hands-on experience in designing power circuits, PCB layout, C/C++ firmware development, and practical substation operations. Seeking an engineering internship in embedded hardware, PCB design, or power electronics.

Education

MDU Rohtak — **Rohtak, India**
B.Tech in Electrical Engineering (2nd Year), 2nd Year — *Present*
Coursework in Power Systems, Control Systems, Embedded Architectures, and Electrical Machines.

Govt Polytechnic College Jammu — **Jammu, India**
Diploma in Electrical Engineering (3-Year Program), 3-Year Diploma — *Completed*
Full 3-year diploma curriculum in Electrical Machines, Circuit Analysis, Power Systems, and Electronics Laboratories.

Bal Bharti Public High School — **Jammu, India**
Secondary School Certificate (10th) — *Completed*
Foundational education in Physics, Mathematics, Chemistry, and Computer Science.

Technical Skills

Hardware & CAD: Analog/Digital Circuit Design, KiCad EDA, Multisim, PCB Layout, Buck/Boost Converters

Embedded Firmware: C, C++, Arduino Framework, STM32 (Nucleo), ESP32 (Wi-Fi/Bluetooth), Python, MATLAB

Lab & Testing: Digital Oscilloscope, LTspice Simulation, Function Generator, Digital Multimeter

Power Electronics: Solar MPPT Controllers, PWM Motor Control, Smart BMS, Battery Storage Systems

Key Engineering Projects

Student Engineering Project — **Jammu**
Lead Hardware Engineer — *Solar Hybrid Buggy*
Designed and built a solar-electric hybrid vehicle featuring roof-mounted PV panels, dual energy charging, PWM motor speed controller, and real-time battery voltage/current telemetry.

Self-Initiated Project — **Jammu**
Power Storage & BMS Lead — *High-Capacity Battery Bank*
Engineered a multi-cell battery pack with series-parallel topology. Integrated Smart BMS for active cell balancing, overcharge/thermal protection cutoffs, and digital state-of-charge (SOC) telemetry.

Hands-on Project — **Jammu**
Drivetrain Conversion — *EV Bicycle Conversion*
Converted a standard mechanical bicycle into an electric bicycle utilizing a Brushless DC (BLDC) Hub Motor, electronic twist throttle, lithium battery pack, and E-ABS brake power cutoff safety levers.

Experience & Training

Gladni Grid Station (220kV / 132kV) **Jammu, India**
Grid Station Operations *4-Week Internship*
Gained practical field exposure to high-voltage switchgear, SF6 circuit breakers, power transformers, air-break disconnectors, busbar protection schemes, and numerical relay testing.

Certified Renewable Energy Training **Jammu, India**
Solar Power Technologies *4-Week Certification*
Trained in solar PV system design, battery storage sizing, PWM vs. MPPT charge controller selection, and grid-tied/off-grid system integration.